Question (1) Choose the correct answer :-

1- The intergranular corrosion is due to ……

(a)presence of stagnant electrolyte	(c) gradual cooling of alloys
(b)presence of aggressive anion	(d)none of these

2- the corrosion rate increases by ……..

(a) decreasing temperature	(c) increasing pH
(b) increasing moisture	(d) all of these

3- …….. ipass reflects strong protection of metal by passive layer

(a) low	(c) moderate
(b) high	(d) has no effect

4- Galvanic corrosion is a …….. reaction

(a) spontaneous	(c) chemical
(b) simultaneous	(d)non-spontaneous

5- The chemical formula of iron rust brown ppt is ……

(a) $Fe_2O_3.3H_2O$	(c) $Fe(OH)_2$
(b) $FeCl_3$	(d) none of these

6- The suitable corrosion medium for Au is ……..

(a)O_2 rich acidic medium	(c) O_2 rich non acidic medium
(b) O_2 poor non acidic medium	(d) none of these

7- Erosion corrosion affected by……

(a) high turbulent	(c) suspended solids
(b) PH	(d) both (a), (c)

8- To control corrosion by changing environment, we should……

(a)remove O_2	(c) increase temperature
(b) adding complexing agents	(d) both (a), (b)

9- In sacrificial anode method, the metal to be protected is…….

(a) more active	(c) less active
(b) the same metal	(d) none of these

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10- The final step in the electrochemical reduction of hydrogen is …….

(a)electrochemical reaction	(c) mass transfer from bulk to surface
(b) mass transfer from surface to bulk	(d) none of these

11- The following factor(s) would affect corrosion rate .

(a) Surface area	(c) Temperature
(b) Humidity	(d) All of these

12- The pitting corrosion is due to ……

(a)presence of stagnant electrolyte	(c) gradual cooling of alloys
(b)presence of aggressive anion	(d)none of these

13- In pitting corrosion, pits grow downward due to .....

(a)the gravity	(c) the stagnant water
(b)the aggressive anion	(d)none of these

14- Which of the following would minimize the intergranular corrosion of stainless steel

(a) decrease carbon content	(c) gradual cooling of alloys
(b)increase the grain size	(d)none of these

15- In differential aeration cell the part of metal in high oxygen concentration tend to be .....while the part of metal in low oxygen concentration tend to be .....

(a)anode, cathode	(c)none of these
(b)cathode, anode	

16- Coating with a more active metal to protect another metal is called……..

(a) impressed current method	(c) cathodic coating method.
(b) anodic coating method.	

17- The ratio between the density of the fuel to the density of water is known as .....

(a) fuel density	(c)specific gravity
(b)viscosity	(d) None of the above

18- In adsorption mechanism of pitting corrosion the second step is.....

(a) migration of the anion through the passive layer	(c) adsorption of Cl^-
(b) reaction with the passive layer	(d) pit formation

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19- In any fuel, presence of .....results in pollution and corrosive environment.

(a) paraffin	(c) sulphur
(b) hydrogen	(d) oxygen

20- High surface tension means ……..

(a)high cohesive force	(c) high adhesive force
(b) good ignition	(d) none of these

21- Among the properties of high quality coating is .....

(a) high porosity	(c) non passive
(b) good adherent	(d) all the previous

22- Low specific gravity means…….

(a) low % of paraffins	(c) good ignition
(b) incomplete combustion	(d) both (a), (b)

23- The shift of the metal potential into a range where passivation occurs is called ……..

(a) Inhibition	(c) metallic coating
(b) anodic protection	(d) none of the above

24- ……cathode +……..anode = corrosion catastrophe

(a) small, large	(c) large, small
(b) large, large	(d) small, small

25- Which of the following involves an electrolytic cell……..

(a) electroplating	(c) vitreous enamel
(b) chemical plating	(d) all of these

26- Stray current corrosion depends on ……

(a) pH of the medium	(c) O_2 concentration
(b) intensity of stray current	(d) all of these

27- ……..is thickening of naturally occurring oxide passive layer

(a) Phosphating	(c) Hot dip coating
(b) Anodizing	(d) Metal spray

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28- Which of the following can be used for cathodic coating of Cu

(a) Cu	(c) Ag
(b) Fe	(d) all of these

29- When stainless steel is kept at 700°C for a period of time, it will suffer from ……..….

(a) pitting corrosion	(c) galvanic corrosion
(b) crevice corrosion	(d) intergranular corrosion

30- As the area of cathode/ anode ratio decrease , the galvanic corrosion.....

(a) increase	(c) decrease
(b) may increase or decrease	(d) will not change

31- Which of the following would have the highest calorific value?

(a) carbon	(c) Sulphur
(b) hydrogen	(d) they are all the same

32- Polarization is..... .....the rate of metallic corrosion.

(a) directly proportional to	(c) has no effect on
(b) inversely proportional to	(d) might be b or c

33- For a piece of iron, which of the following is more corrosive?

(a) NaOH solution	(c) tap water
(b) HCl solution	(d) they are all the same.

34- Self ignition temperature for a given fuel is always .....flash point

(a) greater than	(c) lower than
(b)the same as	(d) might be lower or greater than

35- Presence of sulphur in fuel is associated with.....corrosive environment

(a) acidic	(c) neutal
(b) basic	(d) none of these

36- Which of the following fuels show good ignition? Fuels with…….

(a) high surface tension	(c) High paraffin %
(b) high specific gravity	(d) both **a** and **c**

37- In concentration cell with steel electrodes, the aerated part is……

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(a) passivates	(c) corroded
(b) not affected	(d) none of these

38- For a surface exposed to pitting corrosion, a galvanic cell is produced where the metal inside the pit acts as……..

(a) anode	(c) passive layer
(b) cathode	(d) none of these

39- Changing the metal potential to more cathodic values is called ……..

(a) anodic protection	(c) galvanic corrosion
(b) crevice corrosion	(d) cathodic protection

40- Which of the following is more protective for anodic coating of steel.

(a) Ag	(c) Mg
(b) Fe	(d) all of these

Complete :

1. To avoid ..... we must avoid excessive stress on metals.
2. ....., ...increase rate of erosion corrosion.
3. A passive layer should be ……., & …….
4. Low % of aromatic hydrocarbons in the fuel leads to …….
5. Fuels of ……..specific gravity would have good ignition.
6. As viscosity of fuel increase, its quality ……
7. As C/H ratio increase in fuel, its quality …….
8. ..... is the determining factor in selecting lubricants.

State true or false:

1. Presence of suspended materials in stream flow decrease rate of erosion corrosion.
2. Anodizing is a suitable method for protection of Al.
3. Hot- dip method is suitable for galvanization of steel sheets and wires.
4. Impressed current method involves galvanic cell while sacrificial anode method involves electrolytic cell.
5. Coating steel with a tick layer of cement would decrease te pH value and increase corrosion.
6. As ash content increase in fuel, its quality increase.
7. As paraffin % increase, specific gravity increase.
8. High calorific value is the sum of low calorific value and the amount of heat consumed for evaporation of produced water.
9. a given fuel, flash point is greater than spontaneous ignition temperature.

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10. Volatile fuels show lower flash point than non volatile ones.
11. In any fuel , presence of mercaptens and thioethers leads to air pollution.
12. Insufficient oxygen during burning a fuel leads to incomplete combustion of carbon.

Question (3) Write about each of the following

1. The factors affect the pitting corrosion.
2. the impressed current cathodic protection method for corrosion protection of a pipe line.
3. the causes erosion corrosion.
4. factors affect stress corrosion cracking.
5. Factors affect erosion corrosion.
6. Fctors affecting atmospheric corrosion.
7. Prevent corrosion by controlling the environment.
8. Quality of orgain coatings (paints).
9. Factors which control the quality of surface coating.
10. Anodizing.
11. Galvanization of iron.
12. Electroplating
13. Hot dip method for coating.
14.

Give reason:

1- Our environment is considered as a corrosive environment for many metals.
2- The rate of corrosion in (Zn/Fe) galvanic cell is lower than the rate of corrosion in (Zn/Cu) galvanic cell.
3- The decrease in % of Cr at the grain boundaries in stainless steel alloy .
4- Al doesn't corrode in neutral media.
5- Sulphur is undesirable element in fuel- or as sulphur content increase, quality of fuel decrease.
6- Hydrogen is considered as a perfect fuel.
7- Although coating iron by Cu through displacement method is cheap , it is not the best choice method.
8- Hydrogen gas is considered as an excellent fuel.
9- Some metals loss their chemical reactivity under certain conditions.
10- Rate of atmospheric corrosion depends on the surface conditions of the metal.

Compare between each of the following

1- Galvanic cell and electrolytic cell
2- Cathode and anode in galvanic cell
3- Dry corrosion and wet corrosion
4- Concentration polarization and activation polarization
5- Passive metal and active/passive metal (by graph).
6- Inside crevice and outside crevice.
7- Anodic coating & cathodic coatins.
8- Anodic protection & cathodic protection.
9- Sacrificial anodic method and impressed current method for iron protection.
10- Aerated area and deaerated area in concentration cell.
11- Grain center and grain boundaries in stainless steel alloy.
12- Adsorption mechanism and penetration mechanism for pitting corrosion.
13- High calorific value & low calorific value
14- Flash point and spontaneous ignition temperature.
15- Concentration and activation polarizations

Problems:

I- Deduce the combustion equation of
Carbon(complete and incomplete combustion) – Hydrogen – sulfur- Methane- methanol – ethane – ethanol - Propane - propanol

II- The mass of oxygen required to burn completely 1 Kg of C (H, S).
The mass of oxygen required to burn completely 1 Kg of methane (or methanol – ethane – ethanol - Propane – propanol)

III- Deduce the combustion equation of ethanol (C_2H_5OH), then
A- Calculate the theoretical and actual air supply required for 90% combustion of 7kg of it. (the % of excess air is 40%).
B- Calculate the amount of energy released for that amount of fuel.

IV- Calculate the actual air needed to burn 5kg of fuel 30%carbon,50%hydrogen, 8%water , 5% sulfer,2% ash. Then calculate the energy released, if the combustion 80% and 40% excess air.

V- Calculate the Actual air supply required to burn 10 kg of a fuel that's (40% methanol CH_3OH, and 60% ethanol C_2H_5OH); if the excess air is 20% of the theoretical amount?

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Question(5) Put right or wrong then correct the wrong:

1-Presence of small amounts of moisture could change the corrosion effect.
2-the anodic reaction is any corrosion reaction has the following general form: Mn+ +ne M ( )
3- erosion alone is a chemical or electreochemical process ( )
4-In activation polarization the diffusion step is the controlling step
5-Cu and Pb that are less reactive than steel will provide the required protection
6-Units of kinematic viscosity is cm^2/sec ( )
7- At spontaneous ignition temperature the fuel starts to ignite with flame in presence of air ( )
8- High temperature affect positively on corrosion ( )
9- In polarization metal loss its chemical reactivity and become inert
10-Thickness is factor affect painting quality ( )

Table 16.1 The Standard emf Series

	Electrode Reaction	*Standard Electrode Potential, V^0(V)*
	$Au^{3+} + 3e^- \longrightarrow Au$	+1.420
↑	$O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$	+1.229
	$Pt^{2+} + 2e^- \longrightarrow Pt$	~+1.2
	$Ag^+ + e^- \longrightarrow Ag$	+0.800
ncreasingly inert	$Fe^{3+} + e^- \longrightarrow Fe^{2+}$	+0.771
(cathodic)	$O_2 + 2H_2O + 4e^- \longrightarrow 4(OH^-)$	+0.401
	$Cu^{2+} + 2e^- \longrightarrow Cu$	+0.340
	$2H^+ + 2e^- \longrightarrow H_2$	0.000
	$Pb^{2+} + 2e^- \longrightarrow Pb$	−0.126
	$Sn^{2+} + 2e^- \longrightarrow Sn$	−0.136
	$Ni^{2+} + 2e^- \longrightarrow Ni$	−0.250
	$Co^{2+} + 2e^- \longrightarrow Co$	−0.277
	$Cd^{2+} + 2e^- \longrightarrow Cd$	−0.403
	$Fe^{2+} + 2e^- \longrightarrow Fe$	−0.440
ncreasingly active	$Cr^{3+} + 3e^- \longrightarrow Cr$	−0.744
(anodic)	$Zn^{2+} + 2e^- \longrightarrow Zn$	−0.763
	$Al^{3+} + 3e^- \longrightarrow Al$	−1.662
↓	$Mg^{2+} + 2e^- \longrightarrow Mg$	−2.363
	$Na^+ + e^- \longrightarrow Na$	−2.714
	$K^+ + e^- \longrightarrow K$	−2.924

Best ☺ nes